

Test Case 1: User register account and log in with valid and invalid credentials

Description: User shall be able to register for an account and log into their account with valid and invalid credentials

Test Description	Expected Result	Test Result
User registers for an account	Account successfully registered	Account successfully registered
User logs in with valid credentials	User successfully logged in	User successfully logged in
User logs in with invalid credentials	Login fails with error message “Invalid credentials”	Login fails with error message “Invalid credentials”
User attempts to log in without filling in the username, password, or both	System prompts the user to fill in the missing fields before proceeding	System prompts the user to fill in the missing fields before proceeding

Test Case 2: User information, change password and log out

Description: User shall be able to update their profile information and preference in HDB Listings, change password for their account and log out.

Test Description	Expected Result	Test Result
User updates profile information	User can update their User information and preferences, and successfully save the changes	User can update their User information and preferences, and successfully save the changes
User views profile information	User can see their account username, email, and saved information and preferences	User can see their account username, email, and saved information and preferences
User changes account password	User can change their password by entering the current password and a new	User can change their password by entering the current password and a new

	password	password
User logs out	User is successfully logged out and redirected to the frontpage	User is successfully logged out and redirected to the frontpage

Test Case 3: Homepage

Description: The system shall display all available functions on the homepage that assist the user in filtering and finding their desired HDB listings.

Test Description	Expected Result	Test Result
User click on all the available functions: View all Flats View Bookmarked Flats Go to HDBFinder! Featured neighbourhood listings Account User Information Log out	Each click successfully routes user to the appropriate section/page	Each click successfully routes user to the appropriate section/page

Test Case 4: Bookmark Listings

Description: User shall be able to save desirable listings for future viewing and comparison in the Bookmarked Listings section, as well as delete them when no longer needed.

Test Description	Expected Result	Test Result
User (Logged in) can add a listing to Bookmarked Listing	Listing is successfully saved and appears in the user's Bookmarked Listings	Listing is successfully saved and appears in the user's Bookmarked Listings
User (Logged in) can view Bookmarked Listing	All of user's saved listings are displayed	All of user's saved listings are displayed
User (Logged in) can remove a listing from Bookmarked Listing	Listing is removed and no longer appears in user's Bookmarked Listing	Listing is removed and no longer appears in user's Bookmarked Listing
User (Guest) access any of the bookmark functions	System deny access, prompts user to log in	System deny access, prompts user to log in

Test Case 5: Featured Listings and Listings Overview

Description: The system shall display featured HDB listings based on their preferences in the homepage. Each listing will display brief details, and the user can view more information by clicking on the listing.

Test Description	Expected Result	Test Result
System shall feature HDB Listings based on User's preference	Featured HDB listings are displayed to User based on their preference	Featured HDB listings are displayed to User based on their preference
HDB Listing overview will display the following to the user: neighbourhood name, flat type, cost price and	User can see the brief details of each HDB listing	User can see the brief details of each HDB listing

affordability rating		
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Test Case 6: HDB Listing detailed information with HDB Amenities Display

Description: The system shall display the full details of the selected HDB listing, including nearby amenities. The user shall also be able to view these amenities displayed on a map.

Test Description	Expected Result	Test Result
User clicked on a desirable listing to view the full details of it.	The system shall display the following attributes for each HDB listing: neighbourhood, flat type, cost price, block, street, storey, floor area, model, lease commencement date, remaining lease, and listing posted month. The system displays all listed attributes accurately with corresponding values, and loads a map showing nearby amenities within a 1 km radius.	The system displayed all information and the nearby amenities on a map to the user.
User filtered the HDB amenities display to view specific categories such as transport, schools, clinics/polyclinics, supermarkets, hawker centres, parks, libraries, or community clubs nearby.	The system shall display the selected types of amenities around the HDB unit. The user shall be able to explore the area by clicking and dragging on the map.	The system shall display the selected types of amenities around the HDB unit. The user shall be able to explore the area by clicking and dragging on the map.

Test Case 7: View all Flats filtering

Description: The system shall allow the user to filter HDB listings by the following attributes, which may be used individually or in combination: neighbourhood, price range, and flat type. The system shall display the HDB Listings based on the user's filtering.

Test Description	Expected Result	Test Result
Add "Bishan" to Town Set "Price range" to \$300 000 - \$350 000 Set "Number of rooms" to 3 ROOM Click "Find Flats"	HDB units are displayed accurately according to the selected filters.	More than 20 HDB units are displayed according to the applied filters.
Click "Clear town"	The displayed listings are reset, showing the initial flats with no filters applied..	More than 20 HDB units are displayed with varying towns, flat types, and prices.
Set "Price range" to \$300 000 - \$350 000 Click "Find Flats"	HDB units are displayed accurately according to the selected filters.	More than 20 HDB units are displayed, including a mix of 3-room and 4-room flats from different towns.
Add "Bukit Merah" to Town Click "Find Flats"	HDB units are displayed accurately according to the selected filters.	More than 20 HDB units are displayed with a variety of flat types and prices.
Set "Number of rooms" to 3 ROOM Click "Find Flats"	HDB units are displayed accurately according to the selected filters.	More than 20 HDB units are displayed with a mix of towns and prices.

Test Case 8: HDBFinder filtering

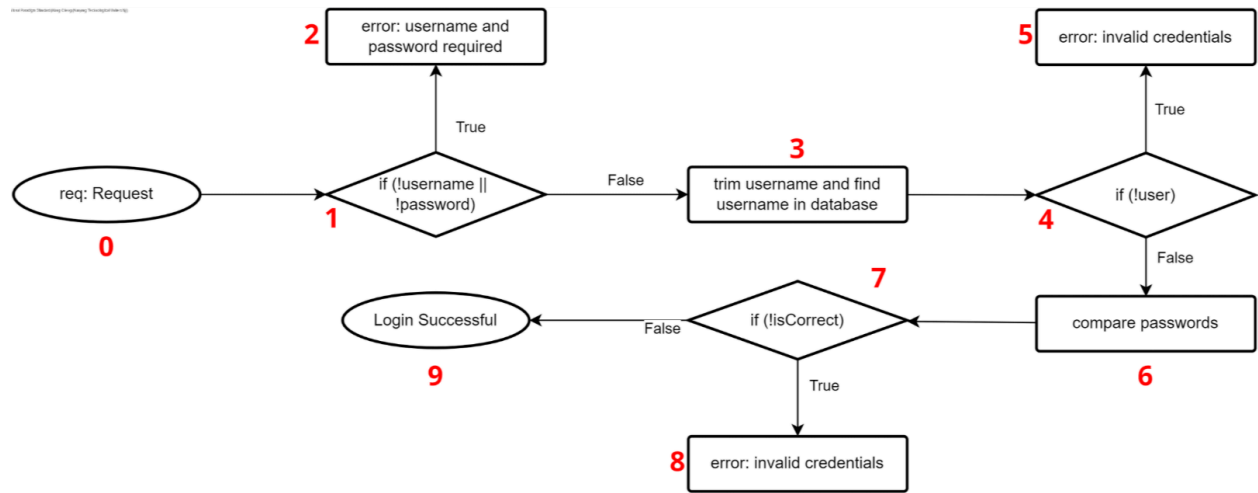
Description: The system shall allow the user to assign a priority rating (on a scale of 1 to 10) to the following attributes: distance to MRT, distance to schools, distance to medical facilities, and affordability rating. The system will display the HDB Listings based on the user's list of priority for amenities.

Test Description	Expected Result	Test Result
Add "Bishan" to town Choose 5 Room Flat Type Set "Distance to MRT" to priority 10 Set other criteria to priority 0 Click "Find Flats"	HDB units are displayed and sorted from low to high according to distance to MRT	5 Room HDB units are displayed and sorted from low to high according to distance to MRT.
Add "Bishan" to town Choose 4 Room Flat Type Set "Distance to MRT" to priority 10 Set "Distance to Schools" to priority 5 Set other criteria to priority 0 Click "Find Flats"	HDB units are displayed and sorted from low to high according to distance to MRT. Among those units with the same distance to MRT, they are sorted according to distance to schools from low to high	4 Room HDB units are displayed and sorted from low to high according to distance to MRT. Among those units with the same distance to MRT, they are sorted according to distance to schools from low to high.
Add "Bishan" to town Set Affordability to 8. Set other criteria to priority 0	All flats in Bishan that have an affordability score of 8 and above are included.	All flats in Bishan are sorted based on affordability.
Add "Bukit Timah" to town Set the 4 criteria to arbitrary values between	HDB units are displayed and sorted from high to low according to the computed score.	HDB units are displayed and sorted from high to low according to the computed score.

0 and 10 Click “Find Flats”		
Add three towns Search and add another town	Cannot add any more town	Cannot add any more town
Add “Bukit Timah” Search for “Bukit Timah”	“Bukit Timah” does not appear in the drop-down list, and hence cannot be added again	“Bukit Timah” does not appear in the drop-down list, and hence cannot be added again
Click “View details”	HDB unit details are displayed	HDB unit details are displayed

White Box Testing – Basis Path Testing

Control Flow Graph of Login Process



Compute Cyclomatic Complexity

$$CC = |\text{decisionPoint}| + 1 = 3 + 1 = 4$$

Basis Paths

We conduct tests for each of the paths, so that we can achieve 100% coverage of all possible branches.

Basis Paths	Description	Test Input	Expected Result	Test Result

0, 1, 3, 4, 6, 7, 9	baseline path, login successful	Enter an existing username Enter the correct password Click “Login”	Login Successful	Login Successful
0, 1, 2	username or password is missing	Enter an existing username Leave password blank Click “Login”	The System displays “Please fill out this field”	The System displays “Please fill out this field”
0, 1, 3, 4, 5	user does not exist	Enter a non-existing username Enter password as “11111111” Click “Login”	The System displays “invalid credentials”	The System displays “invalid credentials”
0, 1, 3, 4, 6, 7, 8	password is incorrect	Enter an existing username Enter a wrong password Click “Login”	The System displays “invalid credentials”	The System displays “invalid credentials”